

VibroCushion: Design for Inclusive Connectedness between Deaf and Hearing in Mindfulness Practice

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Abstract

We present VibroCushion, a vibrotactile system designed to support inclusive mindfulness practices for Deaf and hearing individuals. By delivering vibrotactile stimuli, VibroCushion enhances meditation by improving body awareness and fostering connectedness. Through co-design workshops with Deaf novices, hearing instructors, and Korean Sign Language (KSL) interpreters, the system was iteratively developed to meet the specific needs of Deaf participants. Mindfulness sessions demonstrated that shared vibrotactile stimuli promoted focus, immersion, and a sense of connection between Deaf participants and hearing instructors. Furthermore, VibroCushion shows promise as an assistive tool for emotional communication and inclusive group practices, addressing sensory and social barriers. Our study highlights the potential of VibroCushion to enhance mindfulness and foster meaningful connections in diverse communities, contributing to the broader discourse on inclusive design and haptic interaction.

CCS Concepts

• **Human-centered computing** → Collaborative and social computing; Collaborative and social computing devices.

Keywords

Deaf, Mindfulness, Meditation, Vibrotactile, Connectedness, Assistive Technology

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1 Introduction

Mindfulness involves a non-judgmental focus on the present moment, while meditation is the practice of cultivating mindfulness to enhance present-moment awareness [33]. Recently, mindfulness has garnered significant attention in Human-Computer Interaction (HCI) research [30–32], with a growing emphasis on designing technologies that support meditation practices. Haptic technologies, for example, harness tactile stimuli to enhance focus and immersion [19, 20, 25, 26] or replicate physical sensations associated with meditation [21]. Similarly, visual modalities, such as transforming physiological signals into visual feedback, have promoted mindful breathing [23, 27]. In addition, multimodal approaches integrating visual and auditory elements attempt to improve breath regulation and mindfulness experiences [28].

Meditation encompasses various practices, including mindfulness-based stress reduction (MBSR) [35, 36], breathing meditation [35], and body scan meditation [33]. Body scan meditation, a form of open monitoring meditation [43], involves directing attention to different body parts to observe emerging sensations. Practitioners sequentially shift their focus to various body regions (e.g., head, feet, knees) while observing the sensations that arise [33]. This practice helps beginners experience the benefits of meditation, such as enhanced interoception, improved concentration, and reduced stress [34]. By focusing on bodily sensations, practitioners develop greater awareness of their present experiences, which is fundamental to cultivating mindfulness skills [42]. Although meditation is often perceived as a solitary practice, group-based mindfulness offers unique advantages, particularly for beginners. Practicing mindfulness in a group setting can strengthen emotional and social bonds, amplify the benefits of mindfulness via social support [16, 17], and promote long-term engagement [18]. Recognizing these advantages, HCI researchers have explored group meditation practices, including studies promoting coherence in live-streamed sessions [29] and enhancing social bonds through shared physiological signals [22, 24]. Sharing embodied experiences, such as vibrotactile stimuli, can foster interpersonal connectedness—the emotional and psychological bonds between individuals [44]—by inducing synchronization and a sense of unity via consistent and repeated vibrations [15].

Despite these advancements, prior research has primarily focused on auditory and visual modalities to support mindfulness practices. Vibrotactile stimuli have often been employed as a supplementary sensory channel rather than a primary modality. Consequently, the unique needs of Deaf participants in guided mindfulness practices remain significantly unexplored. The linguistic and sensory differences between Deaf and hearing participants can limit immersive guided meditation experiences. This limitation underscores the crucial need for inclusive mindfulness designs that specifically address the unique requirements of the Deaf community. To address this limitation, we investigate how vibrotactile stimuli can function as a primary modality for facilitating mindfulness practices for Deaf participants. By focusing the design process on the experiences of Deaf individuals, we attempt to develop inclusive technologies that promote individual mindfulness and interpersonal connectedness in group settings. Accordingly, we explore the following research questions:

- RQ1: How does the physical sensation of vibration impact the overall mindfulness experience of Deaf participants?
- RQ2: Do shared vibrotactile stimuli enhance the connectedness between hearing participants and Deaf beginners during mindfulness practice?

To address these research questions, we developed VibroCushion, a set of vibrotactile cushions designed for body-scan mindfulness practice. We adopted a co-design workshop approach to focus the experiences of Deaf participants at the center of the design process. In co-design workshops, end-users and designers collaboratively share knowledge [37], effectively functioning as co-creators in iterative design processes [38]. This approach is crucial for achieving inclusive designs when designing technologies for Deaf participants [14, 39, 40]. During the workshop, we collaborated with Deaf beginners, hearing meditation instructors, and Korean Sign Language (KSL) interpreters to refine the VibroCushion design and structure the mindfulness session iteratively. Subsequently, we conducted mindfulness sessions with one hearing instructor and six Deaf novices.

Our findings demonstrate that vibrotactile feedback fosters a sense of connectedness between Deaf participants and hearing instructors while enhancing focus and immersion during meditation. These results suggest that VibroCushion has potential as an inclusive tool for fostering connection and emotional communication in group mindfulness practices. This study qualitatively elucidates the experiences of Deaf beginners and contributes to the design of haptic technologies for inclusive mindfulness practices.

2 VibroCushion Design

Because VibroCushion is designed to support novice meditators, we carefully considered the form factor and material choices to minimize potential discomfort or intimidation. The VibroCushion system comprises the Body Cushion and Breath Cushion (Figure 1 - a). Cushions feature a soft-knitted fabric exterior, providing a natural, non-intrusive tactile experience that facilitates easy adoption for first-time users. The cushions are powered by Arduino Mega 2560 microcontrollers, which control the actuators and manage interaction inputs efficiently. The Body Cushion features a novel wearable design that transmits vibrations to multiple body

areas based on the user's wearing configuration. This flexibility allows users to explore different wearing configurations to suit their preferences. In addition, the Body Cushion integrates a real-time, touch-based interface that enables meditation instructors to guide body scan meditation, an explored area in related works. Equipped with capacitive touch input sensors, the Body Cushion allows instructors to activate corresponding vibration actuators on the novice's cushion by simply touching specific areas of their cushion (Figure 1 - d). This interactive design enhances coherence in group meditations by transmitting the instructor's tactile cues to novices in real-time.

The Body Cushion weighs approximately 860 g and comprises a chain of six "thin donut"-shaped small cushions, each measuring 14, 38, and 8 cm in width, length, and height, respectively. The Breath Cushion weighs 520 g, with a maximum width of 10 cm (in the section covering the abdomen) and a length of 50 cm (extending from the shoulder down to the abdomen). As illustrated in Figure 1 - b, it wraps around the body to provide targeted vibrotactile stimuli, enhancing awareness of specific body parts. This design ensures flexible vibration delivery based on the user's chosen configuration, facilitating a personalized meditation experience.

The Breath Cushion was designed based on prior research on cushion-type haptic devices that adopt vibrotactile stimuli to regulate respiratory rhythm [41]. Like existing approaches, it employs different haptic signals for inhalation and exhalation to guide breathing patterns. The Breath Cushion, worn around the chest and abdomen, facilitates breath regulation by providing tactile feedback via expansion and contraction (Figure 1 - c). The cushion employs two 5V DC air pumps (flow rate: ≥ 2.5 LPM each) to inflate a flat, silicon-made pneumatic actuator (Ecoflex 00-20), while a 6V three-way solenoid air valve controls air release. This inflation and deflation mechanism guides users to focus on their breathing pace, promoting mindfulness.

3 Study Methods

This study comprises two phases. First, we conducted a co-design workshop with Deaf novices and meditation experts to create a body scan session and discuss design iterations for VibroCushion. Subsequently, we conducted mindfulness sessions with hearing meditation instructors and Deaf meditation novices, followed by structured interviews.

3.1 Co-Design Workshop

3.1.1 Participants. To ensure that the workshop facilitates understanding meditation practices and the Deaf community, we recruited six participants: two meditation experts, two Deaf novices, and two hearing KSL interpreters. The meditation experts had over 12 and 2 years of instructor experience, respectively. Both Deaf novices had meditated once before. The KSL interpreters, with their deep understanding of the Deaf community, provided valuable perspectives and facilitated communication during the workshop.

3.1.2 Procedure. The workshop comprised four stages: introduction to mindfulness, understanding Deaf experiences, hands-on VibroCushion demonstration, and session design (Figure 2).

After the introduction to mindfulness, Deaf participants shared their past meditation experiences. This discussion expanded into

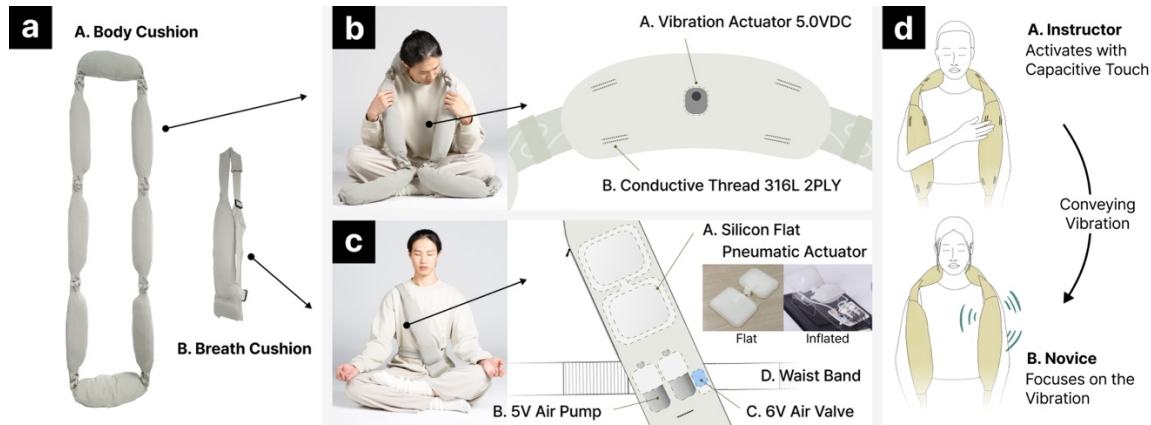


Figure 1: (a) Full view of Body Cushion and Breath Cushion. (b) Body Cushion and its components. (c) Breath Cushion and its components. (d) Instructor conveying vibrations using Body Cushion: Activating vibrations through capacitive touch.

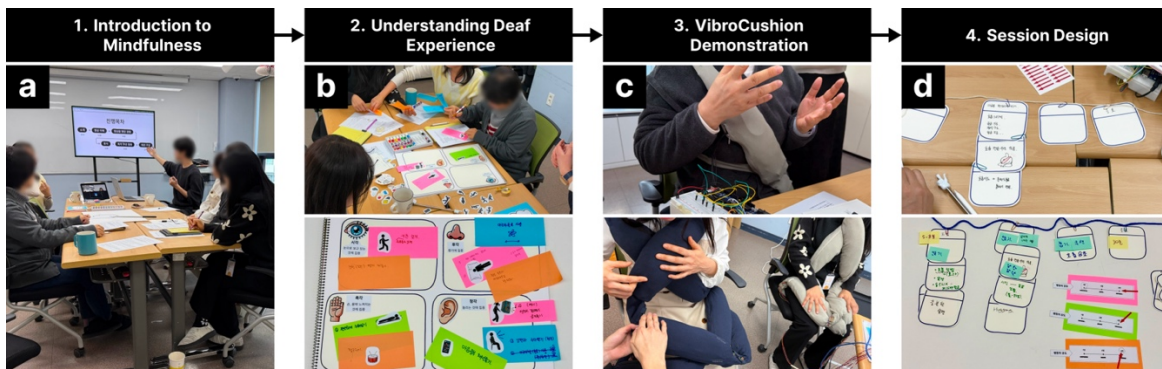


Figure 2: Co-design workshop: (a) Introducing the workshop. (b) Workshop and materials used for the sharing mindfulness activity. (c) Hands-on experience with VibroCushion and brief meditation practice. (d) Workshop material used for session design.

exploring various mindfulness activities in daily life. During the hands-on demonstration, participants discussed optimal cushion arrangements, vibration intensity, and suitable pulsating patterns for vibrotactile feedback.

We utilized various visual materials to facilitate an open discussion among participants. Accordingly, we provided large sticky notes and various images of objects and human figures during discussions on the mindfulness activity, which were then collected on a shared board. Furthermore, prompts were projected on a 48-inch screen throughout the workshop to ensure accessibility. For effective communication, participants were encouraged to speak successively and given an “I Have A Question” sign card for easy interaction.

3.1.3 Results. During the introduction, Deaf participants shared challenges in their past mindfulness experiences, particularly in managing their breathing without auditory cues. One participant reflected, “I struggle to grasp how to breathe properly because I cannot hear. Although hearing people can control their breathing by listening to the sound of their breath, it is difficult for me.” [Deaf 2]. This insight was substantial for hearing instructors, as meditation

sessions are typically designed around breathing patterns. During the subsequent hands-on experience with the Breath Cushion, participants noted that the cushion provided clear and comfortable guidance for following their breath patterns. However, they noted that the conveyed vibrations provided more significant and effective guidance than the pneumatic actuator. One participant shared, “I did not really feel much of the expansion and contraction. I just focused on the vibrations instead.” [Deaf 1]. In response, the Breath Cushion was refined to replace the pneumatic actuator with vibration actuators, ensuring more distinct guidance.

Participants emphasized the importance of salient stimuli for maintaining focus during meditation with the Body Cushion. They highlighted that the clarity and detectability of vibrations were crucial when evaluating three vibration patterns: monotone, slow pulsating, and fast pulsating. One participant remarked, “Whether the vibration is fast or slow does not matter much. What is important is that I can clearly notice where I need to focus.” [Deaf 1]. Participants described feeling uncertain and lost in previous meditation experiences without a clear anchor of attention. Another participant reflected on this: “In the past, we were supposed to close our

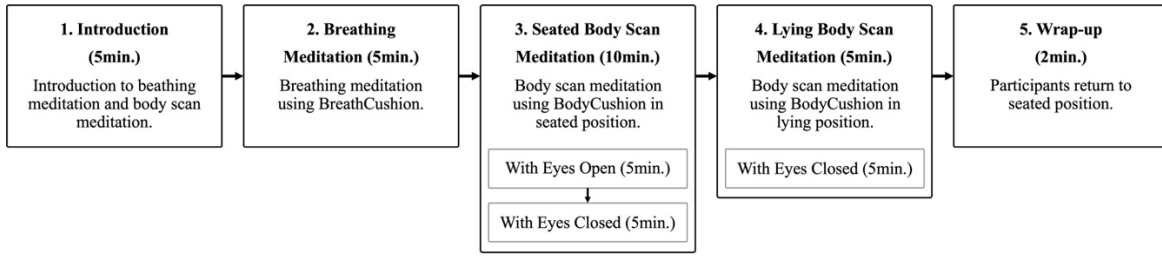


Figure 3: Finalized structure of proposed mindfulness session.

eyes and lie still, but most of us did not understand what to do. This tactile cushion can help Deaf individuals understand meditation more easily.” [Deaf 2]. The meditation experts echoed these insights, emphasizing clarity over precision: “Provided novices can clearly notice the vibration in the general area, it is fine. It does not have to be super precise.” [Meditation Expert 1]. Based on this feedback, we adopted a faster-pulsating vibration pattern for the Body Cushion, as it was easier for participants to detect and follow.

The finalized vibration pattern is as follows. Both cushions employ eccentric rotating mass vibration motors (5V DC, max 9000 RPM) controlled by an Arduino Mega 2560. The Breath Cushion follows an 8-s cycle with four phases: (1) fade-up (1 s), increasing vibration frequency from 0 to 150 Hz; (2) steady-state (3 s) at 150 Hz; (3) fade-down (1 s), reducing back to 0 Hz; and (4) rest (3 s) with no vibration. The Body Cushion follows a 15-s cycle with 12 sequential phases: it starts with a fade-up (1.25 s) to 150 Hz, followed by a steady phase (1.25 s). Subsequently, the frequency alternates between 80 and 150 Hz thrice before a final fade-out phase.

At the end of the workshop, participants collaboratively designed a mindfulness session that supports an inclusive meditation experience between the Deaf participants and the hearing instructors (Figure 2 - d). Participants emphasized the importance of including an introductory phase to familiarize novices with meditation concepts and the VibroCushion system. We integrated a 5-min preparatory exercise into the seated body scan meditation to further support novices. During this exercise, participants placed their hands on the vibrating areas of the cushion to enhance their understanding of the vibration patterns and reduce potential confusion. This preparatory phase helps novices acclimate to focusing on the vibrations, ensuring a smoother and more effective meditation experience. Figure 3 illustrates the finalized set of the mindfulness session derived from the workshop.

Through this workshop, we identified Deaf-specific needs in designing a haptic device for guided mindfulness. Deaf participants lose access to auditory and visual information when their eyes are closed, disorienting their experience. To address this limitation, we established three key design requirements: (1) Continuous haptic feedback, essential for maintaining a sense of guidance throughout the session. Without sustained stimuli, Deaf individuals may experience uncertainty, making it challenging to remain engaged in meditation sessions. (2) Noticeable and unambiguous vibration patterns. In the absence of auditory and visual cues, Deaf novices rely entirely on haptic feedback, prioritizing clarity. Although subtle stimuli are often encouraged in meditation, too indistinct vibrations

may cause confusion and distraction. Hence, vibrations should be sufficiently strong to be effortlessly perceived. (3) Clear mapping of inhalation and exhalation through vibration cues is necessary for breath-guided meditation. Some participants may struggle to perceive their breathing, requiring explicit haptic indicators for inhale and exhale phases. A distinct contrast between these phases enables intuitive breath synchronization.

3.2 Mindfulness Session

3.2.1 Participants. Six Deaf participants with no prior experience in meditation were recruited for the study (4 females and 2 males; age ranges of 25–34: 2 participants, 45–54: 3 participants, and 55–64: 1 participant). Recruitment was conducted via poster recruitment at a Sign Language Interpretation Center located in an inner-city area. The hearing meditation instructor, who also participated in the previous workshop, was a female aged 45–50 with over 12 years of meditation teaching experience.

3.2.2 Procedure. We divided the deaf participants into two groups, with three participants in each group. A hearing meditation instructor facilitated the session for each group. The session procedure was as follows:

1. Participants were briefed on the experimental process and asked to review and sign consent forms.
2. A 30-minute meditation session was conducted (Figure 4) based on the design finalized in the previous workshop.
3. A structured group interview was held immediately after the session to collect qualitative insights. The interview, lasting approximately 30 min, focused on five key topics: physical sensations of the vibrations, impact of vibrations on focus during meditation, sense of connection and trust with the hearing instructor, opinions on shared meditation experiences with other participants, and overall experience with the VibroCushion. The interview comprised 11 predefined questions asked consistently across both groups. All the recordings were fully transcribed, and thematic analysis was employed to analyze the interview data. The first author conducted an initial categorization based on emerging themes, which was subsequently reviewed and discussed with a second author.

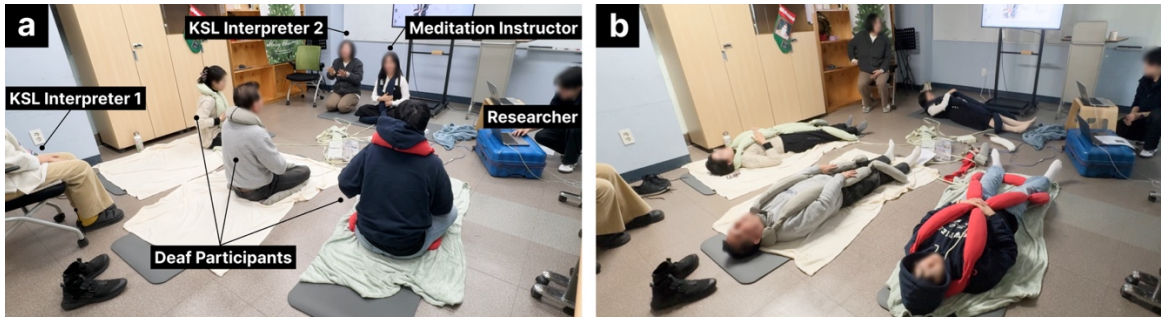


Figure 4: Mindfulness Session: (a) Seated body-scan meditation, (b) Lying body-scan meditation with eyes closed.

4 Findings

4.1 Facilitating Connectedness through Shared Vibrotactile Stimuli

Participants reported that shared vibrotactile stimuli effectively fostered connection and communication with the hearing meditation instructor. For instance, one participant shared, *“It felt like the instructor was calling me, akin to giving me a signal. Normally, I would need to have my eyes open to notice if someone was calling me, but with the vibration, I could immediately tell, ‘Ah, the instructor is signaling me.’”* [P3]. The instructor also commented on the enhanced communication, *“It felt like a shift from verbal instruction to vibration . . . Through vibration, I was able to guide participants.”*

This sense of connection was further reinforced by the real-time nature of the vibrations, which allowed participants to follow the instructor’s guidance intuitively. Vibrations were described as more accessible and immediate than written instructions: *“For me, as a Deaf individual, navigating written sentences can sometimes feel challenging. But with vibrations, it is significantly easier for me to understand. Vibrations feel more intuitive and seem to reach me more directly.”* [P5].

Participants also recognized VibroCushion’s potential for fostering shared mindfulness experiences between Deaf and hearing individuals. One participant wanted to employ the system with hearing family members, stating, *“As it is all about feeling the vibrations, we could all meditate together. We could talk about how it felt to deliver and share the same sensations . . . It would help us create a sense of empathy.”* [P6].

However, participants emphasized the importance of clear communication about vibration patterns before the session to avoid confusion or disruption. For example, one participant felt disconnected when unexpected vibration patterns were introduced: *“During the body scan meditation, the vibration felt different from the breath meditation, and I thought, ‘How am I supposed to breathe with this?’ If I had known the pattern beforehand, I would have felt more in sync with the instructor.”* [P6]. This highlights the need to set expectations to ensure consistency and maintain a sense of connectedness. Although the real-time interactive experience was enhanced by the instructor stroking the cushion, the instructor noted that fidgeting with the device could disrupt meditation when participants have their eyes open. This highlights the need for a

more subtle operation method that maintains the interactive aspect while minimizing the instructor’s movements.

4.2 Comfortable in the Dark while Focusing on Vibrotactile Sensation

Initially, we were concerned that participants might feel discomfort in closing their eyes during meditation, as this had been reported as unfamiliar in group settings during the workshop. However, participants in this study reported no such discomfort. Participants reported that the absence of visual distractions allowed them to immerse fully in the tactile sensations. One participant reflected, *“During today’s meditation, I learned that closing my eyes makes it easier to feel the vibrations and stay focused. It is something I realized after the session.”* [P3].

This immersion solely on vibrotactile sensations helped them focus on specific body parts, preventing distractions. One participant noted, *“Rather than being distracted, I could really focus on the area where I felt the vibration. I did not have to think, ‘Oh, I should focus here.’ I could just ‘focus’ on the vibrations I was feeling in my body.”* [P1].

This bodily focus also fostered a deeper affective connection with their sensations. Participants described how vibrations allowed them to reflect on specific body parts with empathy, enhancing mindfulness and emotional connection with their bodies. One participant shared, *“When the vibration came to my shoulder, I thought, ‘What kind of work did my shoulders do today? . . . They worked so hard.’ It helped me focus on each part of my body, one by one, and I really liked that.”* [P4]. This highlights how vibrations functioned as both a guide for physical focus and a means to cultivate emotional awareness. While this immersion facilitated strong individual focus and connection with the hearing instructor, participants noted a lack of awareness of their peers during the session. They did not perceive other participants’ presence without auditory or visual cues and instead felt more isolated in their individual experiences. One participant stated, *“I feel like I was doing it with the instructor. I did not notice the other participants much.”* [P3]. This suggests that while VibroCushion was highly effective in supporting personal focus and instructor connection, its efficacy in fostering group-level interaction among participants was less pronounced.

5 Discussion

This section discusses how our work extends HCI research on accessibility, along with its design implications, limitations, and future directions.

5.1 Towards an Inclusive Assistive Technology for Deaf and Hearing Communities

Previous studies have demonstrated that transmitting and receiving vibrotactile stimuli between two individuals can foster a sense of connectedness [1], with interaction playing a central role in establishing such connection [15]. Accordingly, our study revealed that the Deaf participants and hearing instructor experienced a sense of intimacy and communication. This suggests that vibrotactile feedback can effectively support the guided meditation experience and create a shared experience between Deaf and hearing individuals. However, no sense of connectedness was reported among novice participants. This is attributable to the design of the VibroCushion, which facilitated a unidirectional interaction from the hearing instructor to the Deaf participants. This underscores the importance of designing systems that enable multi-directional interaction.

In addition, the VibroCushion demonstrates its potential as a collaborative tool accessible to both Deaf and hearing individuals. Previous studies highlight the importance of aligning assistive technologies with users' cultural and linguistic identities to bridge gaps between communities [10] and support communication [11]. One Deaf participant in our study suggested that using VibroCushion with their hearing family members could facilitate mutual understanding, underscoring the potential for nonverbal tactile communication to foster empathy beyond linguistic differences. This indicates that the interaction with vibrotactile stimuli, a sensory modality equally perceivable by both groups, may be promising as an assistive technology for fostering social bonding between Deaf and hearing individuals in non-meditative contexts. Although fostering social connection through shared experience is central to inclusive design, privacy is equally important, particularly in accessibility solutions.

Current visual-based solutions often compromise privacy [1], as accessibility tools such as sign language interpretation and on-screen text expose information to those nearby. In contrast, vibrotactile stimuli provide a private communication channel, as highlighted in this study. This suggests broader applications for haptic interaction beyond mindfulness, particularly enhancing accessibility while preserving privacy for Deaf and hard-of-hearing individuals.

5.2 Potential for Emotional Communication and Assistance

In this study, Deaf novices could focus using vibrotactile stimuli as an anchor to direct their attention to bodily sensations. The VibroCushion demonstrates how guiding body scan meditation via vibrotactile stimuli provides a tangible focal point, facilitating mindfulness and grounding individuals in the present moment. This aligns with prior research in which various haptic stimuli were utilized to support mindfulness practices and enhance body awareness [2–4].

Participants perceived the vibration guidance as calming and relaxing. Meanwhile, earlier studies have noted that vibrations are typically associated with higher alertness [5–7] and are more strongly linked to focus rather than relaxation [8]. This distinction is likely due to the design of the VibroCushion, which delivered vibrotactile stimuli to multiple body areas through a soft, flexible cushion. Previous studies predominantly targeted the hands, with higher vibration detection threshold [9], making the stimuli more prominent and potentially more alerting. Using a soft, knitted fabric in the VibroCushion's construction further contributed to its calming effect. Soft and pliable materials have been demonstrated to evoke a sense of comfort and warmth, potentially reducing the sharpness of vibrotactile stimuli and enhancing their soothing qualities.

Beyond its role in relaxation, our findings also indicate that vibrotactile stimuli can facilitate emotional connection. The VibroCushion fostered an emotional connection with bodily sensations, suggesting the potential of vibrotactile stimuli in facilitating affective experiences. This finding agrees with prior research on affective haptics, which has explored how soft materials can enhance emotional connection by transmitting nuanced vibration patterns [12, 13].

Building on its potential to foster emotional connection, the VibroCushion highlights its potential as a mindfulness tool and a medium for emotional communication and assistance. By leveraging vibrotactile stimuli and soft materials, the proposed VibroCushion could be further developed to convey emotions in remote communication settings. For instance, it could transmit calming vibrations to comfort users during moments of stress or anxiety or even replicate interpersonal connections. This potential to bridge sensory and emotional gaps opens new opportunities for inclusive, haptic-based emotional communication tools, particularly for the Deaf and hard-of-hearing communities.

5.3 Conclusion, Limitations, and Future Work

As a late-breaking work, this study has several limitations that provide opportunities for future research. First, the small sample size limits the generalizability of the findings. While this study provided rich qualitative data from a small group of participants, future work should involve a larger and more diverse participant pool to validate these findings across different cultural and linguistic contexts. Second, although qualitative methods offered valuable insights into the experiences of Deaf participants, we did not incorporate quantitative measures. Future research should combine qualitative and quantitative approaches to evaluate mindfulness, focus, and emotional connection comprehensively. Third, the current VibroCushion design supports unidirectional feedback, primarily enabling a connection between the instructor and participants. Although this approach facilitated individual focus, it did not promote group-level connectedness. Future iterations should explore bi-directional or multi-directional vibrotactile feedback to foster collective mindfulness and group connectedness among participants. In addition, further research could examine the potential of multi-directional shared vibrations to enhance group coherence among hearing participants in mindfulness practices.

This work contributes to the growing body of HCI research by demonstrating the potential of haptic technology to create inclusive, accessible, and socially connected mindfulness experiences. Addressing these limitations in future research could refine VibroCushion, extend its applicability, and elucidate its impact on diverse groups.

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